Lesson 2

Introduction to Agentic AI for Coding

Hands-on with Claude Code on the web

Specify, generate, and verify changes with a coding agent

USAFA Instructor Short Course

Lesson Objectives

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By the end of this lesson you can:

1. Explain how a coding agent differs from a chat LLM, and how your role changes
2. Give an agent your project's context with CLAUDE.md (and AGENTS.md)
3. Specify a change as a verifiable requirement, and get a plan before any code
4. Review an agent's diff and pull request, then verify the result yourself
5. Document GenAI use, and plan how cadets will use agents in your course

Agenda (50 min)

0–15 min	15–45 min	45–50
Concepts	Guided exercises in Claude Code	Review

Prerequisites

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Complete before class:

Lesson 1 repo	Your private copy of **beam-calc** on GitHub, cloned with GitHub Desktop.
Agent account	Claude Pro or higher for Claude Code on the web. Use the account type your organization directs. Other tools: see Agent Tool Options.
GitHub Desktop + VS Code	Installed and signed in (Lesson 1).
MATLAB (optional)	To run the tests yourself. The agent's cloud sandbox has no MATLAB.
Your own project	The repo from Lesson 1's Agent-Ready Checklist, for Exercise 7.

Follow current USAFA guidance on GenAI tools and data handling before you connect a repo to an agent.

Chat LLM vs. Coding Agent

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Chat LLM
Answers in text. You copy, paste, and debug.

→

LLM + tools
Writes and runs code in the chat. You still integrate it.

→

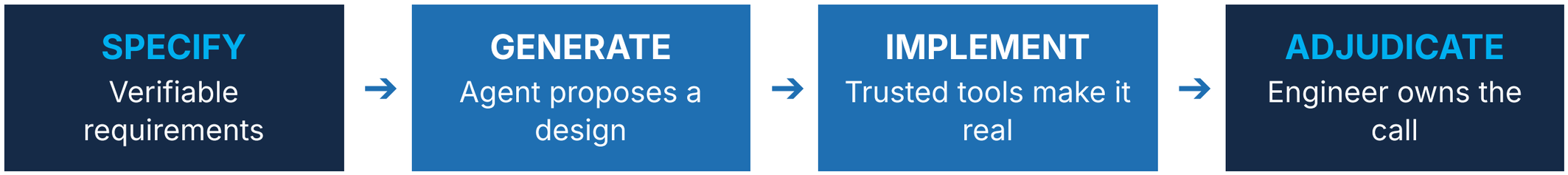
Coding agent
Reads your repo, edits files, runs commands, commits. You review.

	Chat LLM (copy/paste)	Coding agent (Claude Code)
Sees your code	Only what you paste in	Reads the whole repo itself
Edits code	You copy, paste, and merge by hand	Edits files and commits them to a branch
Runs commands	No	Yes, in a cloud sandbox (search, scripts)
Project rules	Re-explained in every chat	Written once in CLAUDE.md, read every session
Your role	Integrator and debugger	Specifier, reviewer, and tester

An agent does more work per prompt. Each prompt needs a clearer specification, and each result needs a closer review.

The Method

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The five workflow steps, for any coding agent:

1. Put your project in a GitHub repo (done in Lesson 1)
2. Open a session in Claude Code, Codex, or another coding agent
3. Connect your repo to the session, with a context file (CLAUDE.md)
4. Work with the agent to generate commits on a branch
5. Adjudicate the agent's work yourself, and iterate

Agent Tool Options

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Tool	Where it runs	Access	Context file
Claude Code on the web (this lesson)	Cloud sandbox, working from your GitHub repo	Claude Pro plan or higher	CLAUDE.md
OpenAI Codex	Cloud tasks from your GitHub repo; also an app, CLI, and VS Code extension	ChatGPT plans; limits vary by plan	AGENTS.md
GitHub Copilot agent mode	VS Code on your laptop; coding agent on GitHub	Copilot Free (limited) or paid plans	.github/ copilot-instructions.md

1. The five workflow steps are the same for every tool.
2. With a local agent, Exercises 2–3 become "open the repo folder in VS Code." The agent edits your working copy; you review and commit in GitHub Desktop.

Plans and limits change often. Check each tool's current plan page, and use the tools your organization approves.

Example: Distributed Load

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SPECIFY

STARTING POINT

Your **beam-calc** repo from Lesson 1. `beam.m` computes cantilever tip deflection for a point load; `tests/testBeam.m` checks it.

REQUIREMENT

beam-calc shall compute the tip deflection for a uniformly distributed load w (N/m) with a function tipDeflectionUDL(w, L, E, I), and beam.m shall print it below the point-load result.

VERIFICATION

Test: a new case in `tests/testBeam.m` matches the hand calculation within 0.1%, and `runtests('tests')` passes in MATLAB. The point-load result is unchanged.

Theory

Point load: $\delta = P L^3 / (3 E I)$

Distributed: $\delta = w L^4 / (8 E I)$

Same beam, P = wL: the distributed result is 3/8 of the point-load result.

```
% Hand check
w = 1000;  L = 2.0;
E = 200e9;
I = 0.05*0.10^3/12;
% δ = 2.40 mm
```

Exercise 1: Add CLAUDE.md

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```
# beam-calc
@AGENTS.md

## About me
- Instructor. Fluent in MATLAB,
  new to coding agents. Explain
  each change before coding.
- Prefer simple, readable MATLAB.

## Environment
- You can't run MATLAB in your
  sandbox. List the checks I
  must run myself.

## Rules
- One function per file.
- Never change an existing
  result or test without asking.
```

EXERCISE

1. In VS Code, create **CLAUDE.md** at the repo root. Adapt the example to you.
2. Line 2, **@AGENTS.md**, imports the rules you wrote in Lesson 1. One set of rules for every agent.
3. Commit to main: *Add CLAUDE.md*. **Push origin**.

Claude Code reads CLAUDE.md at the start of every session. Include what a new colleague would need.

Keep it short. Add a rule each time the agent repeats a mistake.

Exercise 2: Open a Session



The session can read and change your repo. It cannot reach your laptop, your MATLAB license, or any file outside the repo.

EXERCISE

1. Go to **claude.ai/code** and sign in.
2. Connect GitHub when prompted. Install the Claude GitHub App on **your beam-calc repo only**.
3. Accept the **Default** environment.
4. New session: select **beam-calc** and the **main** branch. The session clones the repo and loads CLAUDE.md.

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Exercise 3: Orient the Agent

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EXERCISE

Send an orientation prompt before asking for any change:

YOU

Read CLAUDE.md, AGENTS.md, beam.m, and tests/testBeam.m. Summarize what the code computes, its units, and how it is tested. Don't change any files.

AGENT

beam.m computes the tip deflection of a cantilever under a point load, $\delta = PL^3/(3EI)$, in SI units, and prints it in mm. tests/testBeam.m checks that result; run it with runtests('tests'). I can't run MATLAB here, so I haven't executed the tests.

Check the summary against the code. If the agent misunderstands the code now, its changes will be wrong too.

Exercise 4: Plan Before Code

EXERCISE

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YOU

Implement the requirement: add tipDeflectionUDL(w, L, E, I), $\delta = wL^4/(8EI)$, print it from beam.m, and add a test against the hand calculation (2.40 mm).

Don't write code yet. Give me a plan: files and functions you'll change, and any questions.

AGENT

Plan:

1. New file tipDeflectionUDL.m.
2. beam.m: add w = 1000 N/m; print the result after the point load.
3. tests/testBeam.m: add testUDLHandCalc.

Question: does w act over the full length?

YOU

Full length. Plan approved. Implement it and commit.

Check the plan for: expected files only · real MATLAB functions · questions answered by you, not guessed.

Exercise 5: Review the Diff

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```
claude/udl-deflection · 3 files  +12 −0
@@ tipDeflectionUDL.m (new) @@
+function d = tipDeflectionUDL(w, L, E, I)
+% Tip deflection, uniform load w (N/m)
+d = w*L^4/(8*E*I);        % m
+end
@@ beam.m @@
 fprintf('Tip deflection: %.2f mm\n', ...)
+w = 1000;                  % N/m, uniform
+dW = tipDeflectionUDL(w, L, E, I);
+fprintf('UDL deflection: %.2f mm\n', ...)
@@ tests/testBeam.m @@
+function testUDLHandCalc(tc)
+I = 0.05*0.10^3/12;
+d = tipDeflectionUDL(1000, 2.0, 200e9, I);
+verifyEqual(tc, d, 2.40e-3, 'RelTol', 1e-3)
+end
```

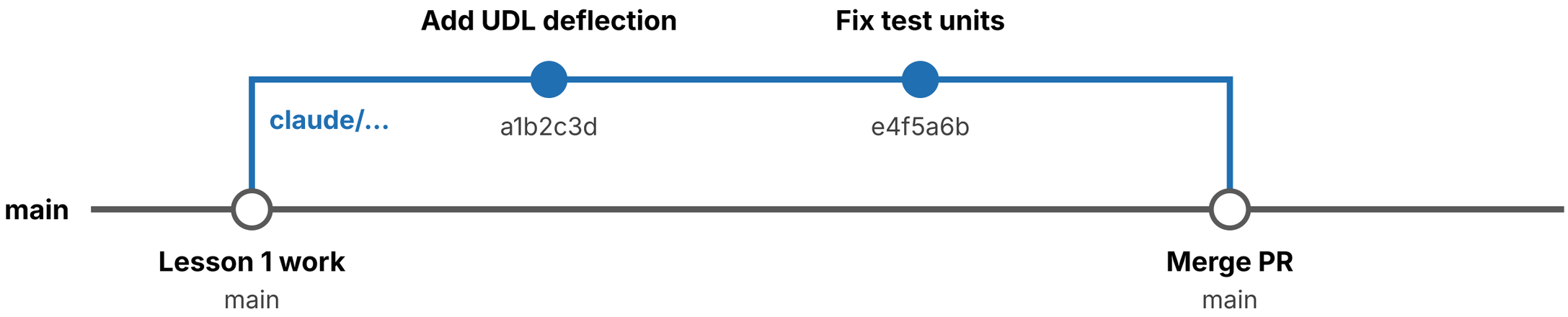
EXERCISE

1. Read every line. Green **+** lines were added; red **−** lines were removed.
2. Ask about anything unclear: *Why RelTol 1e-3?*
3. Select **Create PR** in the session.

You are accountable for every line you merge, as cadets are for every line they submit.

Agent Branches & PRs

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Branch

Each session creates its own claude/... branch and pushes every commit there, including fixes.

main is unchanged until you merge.

Pull request (PR)

1. Select **Create PR** in the session.
2. On GitHub, read **Files changed**.
3. Test the branch yourself (next).
4. Merge only after it passes.

Exercise 6: Test It Yourself

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EXERCISE

1. GitHub Desktop: **Fetch origin**, then switch Current branch to **claude/udl-deflection**.
2. MATLAB: open the repo folder. Run `beam`, then `runtests('tests')`.
3. Check the result against your own hand calculation, not the agent's.

PASS CRITERIA (FROM THE REQUIREMENT)

- [ ] All tests pass, including **testUDLHandCalc**.
- [ ] Hand check: w = 1000 N/m, L = 2.0 m, steel, 50 × 100 mm → **δ = 2.40 mm**.
- [ ] With P = wL, the distributed result is **3/8** of the point-load result.
- [ ] The point-load line prints the same value as on main.

Until you run it, the agent's commit is a proposal.

Iterate & Merge

ITERATE

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If it fails

Report what you observed and paste the output.

YOU

testUDLHandCalc fails: expected 2.4000e-03, got 2.4000e-06. Output below. Find the cause and explain it before changing code.
[paste MATLAB output]

The agent pushes a new commit to the same branch. Fetch, rerun, retest.

If it passes

1. Merge the PR on GitHub.
2. GitHub Desktop: switch to main, **Pull origin**.
3. Start a new session for the next requirement.
4. Add a rule to CLAUDE.md for any mistake the agent made twice.

Common Failure Modes

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Failure	beam-calc example	How you catch it
Wrong formula	w L^4 / (3 E I): the point-load denominator reused	Hand calc; the 3/8 ratio
Unit slip	Section entered in mm in an SI file	Result off by a power of 10
Weakened test	Expected value or tolerance edited until the test passes	Read test diffs line by line
Missing toolbox	Calls a function from a toolbox you don't have	Run it; AGENTS.md toolbox rule
Scope creep	Unrequested "cleanup" of beam.m	Diff touches lines you didn't expect
Unverified claim	"All tests pass"	It can't run MATLAB; you run them

Documenting GenAI Use

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Model the documentation you require of cadets. Follow current USAFA and department guidance for your course.

Keep an **AI_LOG.md** in the repo, with one entry per session:

```
## 2026-10-14 · UDL tip deflection
Session:       claude.ai/code link
Prompts:       orientation, plan, implement, fix test units
Agent made:    commits a1b2c3d, e4f5a6b (PR #2)
I decided:     load acts over the full length
Verified:      runtests passes; hand calc 2.40 mm; 3/8 ratio
Issue found:   test section entered in mm on first try
```

The commit history and PR discussion are also part of the record.

Exercise 7: Your Own Repo

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EXERCISE

1. Open your Agent-Ready repo from Lesson 1.
2. Write one small requirement **and how you'll verify it**, before opening a session.
3. Add CLAUDE.md, or run **/init** and correct it.
4. Orient → plan → implement.
5. Review the PR and test it yourself. Merge only if it passes.
6. Add an entry to AI_LOG.md.

Good first requests

- Add a test for an existing function
- Add input checks with clear errors
- Explain how to run it in the README
- Replace hard-coded values with parameters

Avoid for now

- "Rewrite the whole project"
- Anything you can't check yourself

Cadet Teams & Agents

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An agent session belongs to one account; teammates can't join it. The GitHub repo is the team's shared workspace.

OPTION A

One driver

- One cadet with agent access runs the sessions.
- The others review plans and PRs, and test.
- Rotate the driver each requirement.

OPTION B

Parallel drivers

- Each cadet runs sessions on their own requirement and branch.
- Merge through PRs.
- Don't let two sessions edit the same function.

Grade the process: requirement, plan, PR review, test evidence, AI_LOG.md.
Add every teammate as a repo collaborator. Never share agent account logins.

Troubleshooting

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Symptom	Fix
Repo missing from the session list	Grant the Claude GitHub App access to that repo (GitHub › Settings › Applications).
Agent ignores your rules	Check CLAUDE.md is at the repo root on the branch the session uses. Shorten it.
Agent's branch not in GitHub Desktop	Fetch origin first, then open the Current branch menu.
Agent says tests pass	It can't run MATLAB. Run runtests('tests') yourself.
PR changes lines you didn't ask for	Comment on those lines in the PR, or ask the agent to revert them.
Answers drift in a long session	Merge what works, then start a new session per requirement.

Quick Reference

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To...	Prompt or action
Orient the agent	"Read CLAUDE.md and ... Summarize ... Don't change any files."
Get a plan	"Don't write code yet. Give me a plan: files, functions, and questions."
Approve the plan	"Plan approved. Implement it and commit."
Report a failure	"Observed ...; expected .... Output below. Find the cause and explain it before changing code."
Draft CLAUDE.md	/init in a session, then correct it
Open a PR	Create PR in the session
Test the agent's branch	Fetch origin › Current branch › claude/...
Finish	Merge the PR, switch to main, Pull origin

Knowledge Check

1. What can a coding agent do that a chat LLM can't, and how does that change your role?
2. What belongs in CLAUDE.md, and how does it relate to AGENTS.md?
3. Why ask for a plan before any code is written?
4. The agent reports "All tests pass." What do you do?
5. A PR changes a test's expected value so the test passes. Merge it?

Next: Lesson 3 · Agentic AI for course app development using GitHub Pages.

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